

SAP Convergent Charging (CC 2020/2023) Advanced Architecture & Implementation Guide

Product: SAP Convergent Charging

Module: CC / BRIM

Audience: Senior SAP BRIM Enterprise Architects & Consultants (8+ Years Experience)

Classification: Enterprise Official Technical Handbook

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Section 1: SAP CC Core Server System Architecture & Instances Topology

SAP Convergent Charging (CC) is the ultra-high throughput rating and charging engine of SAP BRIM. The Core Server consists of several specialized instance types operating in a distributed cluster:

- Dispatcher Instances: Manage incoming TCP/HTTP client connections, perform connection pooling, load balancing, and route stateful charging requests to Guider and Rater instances.
- Guider Instances: Maintain the mapping of User Technical Identifiers (UTI) to Subscriber Contracts and partition IDs, ensuring request affinity to the correct partition.
- Rater Instances: Execute dynamic rating and charging trees, perform math calculations, update persistent and transient counters, evaluate pricing logic, and determine charge amounts.
- Updater Instances: Manage subscriber account balances, commit wallet/prepaid balance changes to the database, and execute transactional persistence.
- Bulkloader Instances: Extract charged and consumption items from internal buffers and load them directly into SAP Convergent Invoicing (CI) tables or BART files.
- Taxer Instances: Integrate with internal or external tax calculation engines (Vertex, Avalara, SAP Tax Engine) during real-time charging.

Reference: SAP BRIM Technical Standard - CC / BRIM - Section 1

Section 2: Hazelcast In-Memory Distributed Data Grid (IMDG) Architecture

SAP CC Core Server relies on Hazelcast IMDG for in-memory state replication, high availability, and microsecond response times. Key Hazelcast aspects include:

- Partitioning: Subscriber accounts and active sessions are distributed across partitions (typically 271 or 511 partitions).
- Near Cache: Rater instances maintain local Near Caches for frequently read master data such as Translation Tables, Range Tables, and Charge Plans.
- Cluster Split-Brain Protection: Quorum rules and network heartbeat parameters prevent desynchronization across active-active data center topologies.
- Cache Refresh Commands: Administrators can trigger near-cache invalidation using the CLI tool 'admin.bat -cache -refresh' without restarting Core Server instances.

Reference: SAP BRIM Technical Standard - CC / BRIM - Section 2

Section 3: Master Data Hierarchy: Pricing Catalog, Charge Plans, and Charges

Master data in SAP CC is organized in a strict hierarchical structure:

1. Service Provider: Root entity representing the company code or operating business unit.
2. Pricing Catalog: Container of reusable pricing objects, tables, and charge plans owned by a service provider.
3. Charge Plan: The primary commercial building block exposed to SAP SOM. A charge plan contains one or more Charges and technical interface parameters.
4. Charge: Represents an individual pricing rule (e.g. Monthly Fee, Voice Usage, Data Usage). Each charge contains a Price Plan and a Charging Plan.
5. Price Plan: The rating logic tree that computes the monetary amount based on event properties.
6. Charging Plan: The routing logic tree that determines which subscriber account (prepaid wallet or external postpaid account) pays for the charged amount.

Reference: SAP BRIM Technical Standard - CC / BRIM - Section 3

Section 4: Rating Tree Components & Decision Logic Engineering

Price plans and charging plans are modeled as visual decision trees comprising specific node types:

- Comparators: Number Comparator, String Comparator, Date Comparator. Used to branch execution based on call destination, time of day, customer tier, or event volume.
- Splitters: Time Splitter (splits usage across peak/off-peak windows), Volume Splitter (splits data usage across quota thresholds).
- Tables: Translation Tables (key-value lookups), Range Tables (bracket-based pricing tiers), Tier Tables (multi-dimensional matrices).
- Pricing Macros: Encapsulate reusable calculation sequences across multiple charge plans to enforce standard logic and simplify maintenance.
- Counters: Numerical variables tied to the subscriber contract. Persistent counters are committed to the DB; transient counters reset at session end.

Reference: SAP BRIM Technical Standard - CC / BRIM - Section 4

Section 5: Allowances and Allowance Packages Architecture

Allowances represent customer entitlements to free or discounted consumption (e.g. 10 GB free data, 500 bonus minutes):

- Allowance Definition: Master data template specifying allowance behavior, parameters, and validity lifecycle.
- Allowance Plan: Logic tree defining how an allowance is credited, debited, extended, or expired.
- Allowance Logic in Rating Trees: Rating logic inspects available allowances first. If an allowance exists with remaining balance, the event consumes the allowance balance and generates a zero-rated or discounted billable item. Once exhausted, standard tier rating resumes.
- Shared Allowances: Allowance packages can be shared across multiple provider contracts in a family pooling or corporate account scenario.

Reference: SAP BRIM Technical Standard - CC / BRIM - Section 5

Section 6: Subscriber Accounts, Prepaid Accounts, and User Technical Identifiers (UTI)

Subscriber data links technical network activity to financial accounts:

- Subscriber Account: Created per customer; holds prepaid balances and external account references matching SAP FI-CA Contract Accounts.
- Prepaid Accounts: Virtual wallets supporting real-time credit checks, minimum balance thresholds, and top-up refill plans.
- User Technical Identifier (UTI): Network-level service identifiers such as MSISDN phone numbers, IMSI, IP addresses, MAC addresses, or email addresses. Guider instances use the UTI service identifier to instantly locate the target contract.

Reference: SAP BRIM Technical Standard - CC / BRIM - Section 6

Section 7: Diameter Ro/Gy Online Credit Control & Sy Spending Limits

In telecom and utility environments, SAP CC integrates with 3GPP network nodes using Diameter protocol interfaces:

- Diameter Ro / Gy: Provides session-based charging (CCR-Initial, CCR-Update, CCR-Terminate) with Credit Control Answers (CCA) granting granted service units (GSU).
- Diameter Sy Interface: Communicates policy spending limit information between SAP CC and the Policy and Charging Rules Function (PCRF), notifying when thresholds are breached.
- Event-Based Charging: One-time direct debit events (Direct Debit Requests) for instant content purchases or service activation fees.

Reference: SAP BRIM Technical Standard - CC / BRIM - Section 7

Section 8: Batch Acquisition & Rating Tool (BART) Operations

BART handles offline, asynchronous mass processing of usage records:

- Consumption Data Records (CDRs) and Event Data Records (EDRs) are ingested via batch file loaders.
- BART Server performs deduplication, structural validation, and dispatches batch rating requests to Rater instances.
- Processed items are output as Billable Items directly into Convergent Invoicing staging tables via Bulkloader.

Reference: SAP BRIM Technical Standard - CC / BRIM - Section 8

Section 9: Core Tool, Cockpit, and Administrative CLI Tooling

Administrators and pricing specialists manage SAP CC using dedicated interfaces:

- Core Tool: Rich Java Swing client for creating pricing catalogs, charge plans, rating trees, and subscriber accounts.
- Cockpit: Modern SAP Fiori web interface for monitoring server metrics, managing user permissions, and inspecting system alerts.
- Administrative Scripts: 'setup.bat' (system initialization, schema creation), 'admin.bat' (runtime cluster management, pinging instances, inspecting thread pools, updating licenses), 'config.bat' (exporting/importing XML configuration).

Reference: SAP BRIM Technical Standard - CC / BRIM - Section 9

Section 10: SAP CC Troubleshooting, Latency Optimization, and War Room Best Practices

Key production troubleshooting steps for senior CC architects:

- Diagnosing Rating Latency: Inspect garbage collection logs (G1GC pauses), check Hazelcast near-cache hit ratios, and examine DB connection pool saturation.
- Resolving JCo RFC Timeouts: Increase connection pool size and max connections in 'jco.server.connection_count'.
- Unlocking Stuck Sessions: Use 'admin.bat' commands to release orphaned session reservations caused by abrupt network dropouts.
- Verifying Partition Rebalancing: Ensure no data skew across rater partitions following server instance restart.

Reference: SAP BRIM Technical Standard - CC / BRIM - Section 10